

ECE 657D Neural Networks and Deep Learning Winter 2026

Research Project

Proposal Due Fri Feb 27 by 11:59 p.m.

Final Report Due Thu Apr 2 by 11:59 p.m.

Overview

This project is for **ECE 657D students only**. ECE 493 students will instead complete a paper report.

The research project is a core component of ECE 657D. The project gives you the opportunity to explore a deep learning research question of your choosing in depth. You will identify a research question, situate it within the existing literature, propose and implement an approach, and report your findings.

This is an *individual* project. You are expected to work independently, though you are welcome to discuss ideas at a high level with classmates.

This is a short research project meant to give you a flavour of what academic research is like. For the final report, you are **not** expected to set the new state-of-the-art. The main point is to demonstrate that you have followed a good research *process*, regardless of the outcome. The final report will be four pages (typeset in L^AT_EX, excluding references and appendices) and should describe a short but well-designed project that investigates an interesting research question.

Learning Objectives

By completing this project, you will:

- Formulate a well-scoped research question in deep learning
- Conduct a focused literature review and identify relevant prior work
- Design, implement, and evaluate a deep learning experiment
- Communicate your findings clearly in a short scientific write-up

Deliverables

Proposal (due Feb 27)

- **Format:** 2-page PDF (excluding references), using the provided L^AT_EX template
- **Content:** Address all eight adapted Heilmeier questions (see below). At the proposal stage, some answers may be preliminary—that is expected. The proposal should demonstrate that you have (1) identified a clear question, (2) done initial literature review, and (3) have a feasible plan.
- **Submission:** Submit as a single PDF to **Crowdmark**

Final Report (due Apr 2)

- **Write-up:** 4-page PDF (excluding references and appendices), using the provided L^AT_EX template. This is an expanded and updated version of your proposal that includes your results and analysis.
- **Code and data:** Your report should include a link to a repository containing your code and instructions for obtaining any relevant data. Using your course-provided GitLab repo (<http://git.uwaterloo.ca/deep-learning-w26/grad/<username>>) is strongly recommended.
- **Submission:** Submit as a single PDF to **Crowdmark**

Stage 1: The Project Proposal

Your proposal will be evaluated according to an **adapted Heilmeier Catechism**—a set of questions originally developed by George Heilmeier (former DARPA director) to evaluate research proposals.¹ We have adapted these questions for the context of a graduate-level deep learning course project.

Your proposal should address *all eight* of the following questions. The provided L^AT_EX template includes a section for each.

Q1. What is your research question?

State your research question in plain language. What are you trying to accomplish? A reader with no background in your specific topic should be able to understand what you are investigating.

Q2. What prior work will you compare against, and what are its limitations?

Summarize the most relevant existing approaches. What has been tried before? What are the key limitations or gaps that motivate your work? Cite specific papers.

Q3. What is novel about your proposed approach?

Explain what is new or different about your method. This does not need to be a major research contribution—it could be a novel combination of existing techniques, an application to a new domain, or a systematic empirical comparison that has not been done before.

Q4. Why does this matter? What is the potential impact?

Who would benefit from your results? Why should we care about your research question? Connect your work to broader goals in deep learning or its applications.

Q5. What could go wrong? What are the key assumptions?

Identify the main risks and assumptions underlying your approach. What are the failure modes? Are there assumptions about the data, compute, or method that might not hold?

Q6. What compute and data resources do you need, and how will you obtain them?

Describe your resource plan. What datasets will you use? What compute do you need, and where will you get it? Be specific. See the *Compute Resources* section below for guidance.

¹See <https://www.darpa.mil/about/heilmeier-catechism> for the original Heilmeier Catechism.

Q7. Provide a week-by-week plan for the ~ 5 weeks between proposal and final submission.

Break your project into concrete weekly milestones. What will you accomplish each week?

Q8. What are your milestones, and how will you measure success?

Define quantitative and/or qualitative metrics you will use to evaluate your approach. What constitutes a “successful” project? What baselines will you compare against?

Stage 2: The Final Report

Your final report uses the same L^AT_EX template as the proposal. Remove the eight Heilmeier questions from the template and replace them with your own sections. Your report should include the following sections: Abstract, Introduction, Related Work, Method, Experiments, and Discussion. A Conclusion section is optional but recommended. Appendices may be used for any extra details (e.g., additional figures, derivations, or hyperparameter settings) that do not fit within the four-page limit.

Compute Resources

This course does not provide dedicated compute. Here are some options:

- **Google Colab** (free tier): Provides limited GPU access, sufficient for many projects
- **Kaggle Notebooks**: Free GPU access (30 hrs/week)
- **UW campus clusters**: If you have access through your research group
- **Your own hardware**: If you have a GPU-equipped machine
- **CPU only**: As we saw in A1, backpropagation works fine on CPUs—training is just slower. If GPU hardware is not available, consider working with smaller models and smaller datasets. Tabular data, for instance, is typically much less memory-intensive than image or text data and can be a good fit for CPU-based projects.

Your project scope should be calibrated to the compute you have available. A well-designed small-scale experiment is preferred over an ambitious but infeasible large-scale one. You will not be penalized for lack of access to external GPU resources; however, even for small-scale projects, you will be expected to demonstrate adherence to a good research process.

Grading

The proposal is worth **20 marks** and the final report is worth **80 marks**.

Grading will assess:

- **Clarity**: Is the writing clear, concise, and well-organized?
- **Technical depth**: Does the project demonstrate understanding of the relevant methods?
- **Soundness**: Are the experiments well-designed and the conclusions supported by the results?
- **Scope**: Is the project appropriately scoped for the time and resources available?

- **Novelty:** Does the project offer some original contribution, however small?

Policies

Use of AI Tools

You may use AI coding assistants (e.g., GitHub Copilot, ChatGPT) to help with implementation, but you must:

- Disclose any AI tools used in your write-up (e.g., in an acknowledgments section)
- Understand and be able to explain all code you submit
- Write your project text yourself: directly using AI to generate your prose is not acceptable, although it's fine to use AI (Grammarly, ChatGPT, etc.) to proofread your own writing.